Sequences and Basic Series Convergence Tests

Convergence of sequences

1 Determine whether the sequence $a_n = \frac{3n+1}{n}$ converges, and if so, find its limit.

2 Determine whether the sequence $a_n = (-1)^n$ converges.

Geometric series

3 Determine whether $\sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$ converges, and if so, find its sum.

4 Determine whether $\sum_{n=1}^{\infty} 2^n$ converges.

The nth-term (divergence) test

5 Use the nth-term test to show $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges.

The integral test

6 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

7 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges (the harmonic series).

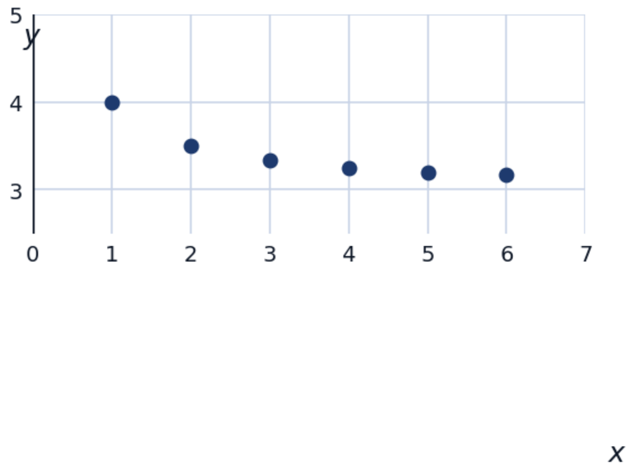
The comparison and limit comparison tests

8 Use the Direct Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges, comparing to $\sum \frac{1}{n^2}$.

9 Use the Limit Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{2n}{n^2+3}$ converges, comparing to $\sum \frac{1}{n}$.

Visualizing sequence convergence

10 Plot the first 6 terms of $a_n = \frac{3n+1}{n}$ (from the earlier question) and describe how the points behave as n increases.



Monotonic and bounded sequences

11 Determine whether $a_n = \frac{n}{n+1}$ is monotonic (always increasing or always decreasing), and find an upper bound.

12 State the Monotonic Sequence Theorem, and use it to conclude that $a_n = \frac{n}{n+1}$ converges.

The p-series test

13 State the p-series test for $\sum_{n=1}^{\infty} \frac{1}{n^p}$, and classify convergence for $p = 1$, $p = \frac{3}{2}$, and $p = \frac{1}{2}$.

14 Use the p-series test (after simplifying) to determine whether $\sum_{n=1}^{\infty} \frac{5}{n^{0.9}}$ converges.

Series review: choosing the right test

15 For each series, name the most efficient test to apply (do not fully evaluate):

a $\sum \frac{1}{5^n}$

b $\sum \frac{1}{n^3}$

c $\sum \frac{n+1}{2n+3}$

d $\sum \frac{1}{n^2+2n}$

Sequences defined recursively

16 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2, a_3, and a_4, and conjecture the limit if the sequence converges.

17 Verify that if a recursively defined sequence $a_{n+1} = \frac{a_n + 6}{2}$ converges to a limit L, then L must satisfy $L = 6$.

Geometric series with algebraic manipulation

18 Find the sum of $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n$ (note the series starts at $n = 2$, not $n = 0$).

19 Express the repeating decimal $0.\overline{45}$ as a geometric series and find its exact fractional value.

Telescoping series

20 Evaluate the telescoping series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$ by examining its partial sums.